

M003 Week 6 — English Worksheet

Topic: Building Solid Shapes with the Builder · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you use the **builder** — an invisible helper that moves through the world. The pattern is always the same: move the builder to the start, **place a mark**, move along **x**, **y**, or **z**, then **trace** or **fill** between the mark and where the builder is now.

1 · Predict 🧠

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
move builder to (0, 0, 0)
place mark
move builder along x by 5
fill from mark with stone
```

The builder marks, then moves along x only. What shape appears — a line, a wall, or a box?


```
move builder to (0, 0, 0)
place mark
move builder along x by 5
move builder along y by 3
fill from mark with stone
```

Now the builder also moves up along y. What shape does **fill** make this time? How tall is it?


```
move builder to (0, 0, 0)
place mark
move builder along x by 5
move builder along z by 5
move builder along y by 3
fill from mark with stone
```

The builder moves along all three axes — x, z, and y. What shape do you get now? Is it flat or solid?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should build a stone line from **(0, 0, 0)** to **(5, 0, 0)**. Right now only one block appears.

```
move builder to (0, 0, 0)
move builder along x by 5
place mark
fill from mark with stone
```

Hint: the mark must go down **before** the builder moves, not after.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should trace a wall that **starts at the corner (0, 0, 0)**. Right now the wall appears wherever the builder happened to be standing.

```
place mark
move builder along x by 6
trace wall with stone
move builder to (0, 0, 0)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should fill a stone wall 5 long and 3 tall. Right now `fill from mark` has nothing to work with.

```
move builder to (0, 0, 0)
move builder along x by 4
move builder along y by 2
fill from mark with stone
```

Hint: read every line — is there a `place mark` anywhere?

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Use the builder to make a **solid shape**: move to a start corner, place a mark, move along at least **two axes**, then fill. When you finish, come back here.

Send a photo or video of your shape, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I moved the builder to ...

I placed the mark before ... because ...

The builder moved along ... and then along ...

The shape I got was ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you show your shape in the world. Talk through it like you are teaching someone who has never seen the builder. Try to use these words: **builder, mark, trace, fill, axis.**

1. Show the empty spot, then run your code so the shape appears.
2. Point at the start corner and say where the mark went down.
3. Say which axes the builder moved along — x, y, or z — and in what order.
4. Say in your own words why the mark must come **before** the builder moves.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.